

MODULE ONE

DEFINITION OF DATA COMMUNICATION AND DIFFERENT EQUIPMENT AND COMPONENTS USED

1.1 DEFINITION OF DATA COMMUNICATION

Data communications is the transmission of encoded information from one point to another by means of electrical or optical transmission systems. Such systems are often called **data communication networks**. A network consists of an interconnection of a number of devices (often referred to as **nodes**) connected by communication links. A node can be a computer, mobile phone, or any other terminal capable of creating, processing, sending and/or receiving encoded information generated by other devices on the network.

For data communication to occur the data processing devices must be part of a data transmission or telecommunication system. Data processing devices rely on the telecommunication system to convert, amplify and send the encoded information or message through a common medium. To illustrate the relationship between Data communications and Telecommunications, we group these two terms together with the aid of computers. Data communication makes use of the computer to process the digital information and relies on telecommunication equipments to deliver the information to remote receiver as shown in Figure 1-1.

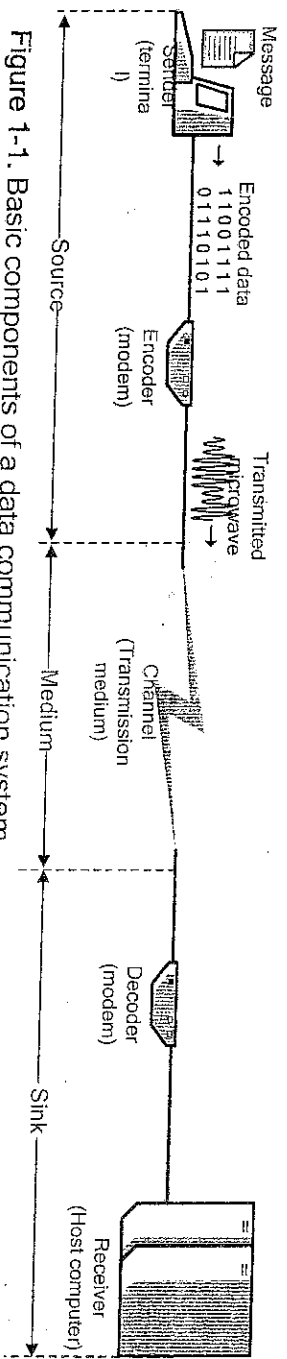


Figure 1-1. Basic components of a data communication system

Encoded information refers to digital (that is discrete) data, which is a sequence of binary digits (series of ones and zeros), called bits. Binary system is used because of the following reasons;

- ⇒ It is simple and universally recognized.
- ⇒ Digital data have greater immunity to noise and interference than analog data.
- ⇒ Digital data can be easily processed and stored.
- ⇒ Digital components and devices that process digital data can easily be miniaturized (reproduced at extremely small sizes) for example micro/nano SIM cards, memory card.
- ⇒ Energy/power requirement for processing and transmitting digital data is extremely low
- ⇒ The advantage of miniaturization and extremely low energy requirement makes data communication generally affordable and popular.

The most common data sources and destinations are portable electronic devices, such as computers, tablets, mobile phones, PDAs, portable media players, digital cameras, video game consoles etc.

Example of data communication system are, Global System for Mobile Communication (GSM) service, Automated Teller Machine (ATM) Service, Computer Networks (LAN, MAN, WAN), Internet Service, Global Positioning System (GPS).

1.2 COMPONENTS OF A DATA COMMUNICATION NETWORK

Data communication network has three basic parts the **source**, the **medium**, and the **sink** (see Figure 1-2). The encoded data (message) is created at the source; the medium is the path through which the message flows; and the sink is the system that receives the message. These components are broken up into five components: Message, Sender, Channel, Receiver, and Protocol.

1. **Message.** The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
2. **Sender.** The sender is the terminal (device) that creates and transmits the encoded data (message). It can be a computer, workstation, mobile phone, video camera, and so on. Generally, because the data message created by the sender is unsuitable for immediate sending, an encoder (modem) is used to transform the message into a form suitable for transmission over the channel.
3. **Channel.** The channel (transmission medium) is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fibre-optic cable, and radio waves.
4. **Receiver.** The receiver is the device that receives the message. It can be a computer, workstation, mobile phone, television etc. However, a decoder (modem) is required to convert the message back into the form that the receiver can recognize.
5. **Protocol.** Protocols are the unseen components that controls and supervises the transmission of the message until it is received at the destination. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without protocols all the devices may be connected but not communicating. Common examples are the Transmission Control Protocol (TCP), and Internet Protocol (IP) used on the Internet.

1.3 NEED FOR DATA COMMUNICATIONS

Data communication networks make the most use of computers to make life easier and to solve human problems in different areas of human life and this includes scientific, research, business, education, medical and social aspects. In general, they offer better and rapid transfer of data among their users and bring the power of computing closer to more users. The aims of most data communication networks are to;

- ⇒ Effect rapid dissemination of information
- ⇒ Reduce the time and effort required to perform various business tasks
- ⇒ Capture business data at its source
- ⇒ Centralize control over business data

- ⇒ Reduce current and future costs of doing business
- ⇒ Support organizational objectives in centralizing and decentralizing computer systems

It is beneficial if computers can exchange their data through the common links between them. This not only reduces document flow, but also saves time. For example some companies use Electronic Data Interchange (EDI) to do business such as placing order and sending invoices.

Retail shops use Point-of-Sale systems to keep and check the inventory level locally, while sales summaries and request can be transmitted to the computer in headquarters for overall turnover.

It provides a way to link hardware, software, and databases among computer systems in different geographical locations and what is more important is that this enhances a company to make better use of its computing resources. For example regional office may send a job, which is too big for its regional computer system to handle, to the headquarters' computer system for processing and sending the result back.

1.4 CLASSIFICATION OF DATA COMMUNICATION SYSTEMS

There are five broad classes of data communication system these are (in order of increasing sophistication): --

- | | | |
|---------------------------------|------------------------|------------------------------|
| (1). Off-line Data Transmission | (2). Remote Batch | (3). On-line Data Collection |
| (4). Enquiry-response Systems | (5). Real-time Systems | |

Note that the classes are compatible with each other; some applications fall into more than one class.

- (1) **Off-line Data Transmission:** this describes a data collection terminal, device or peripheral that is disconnected or is functions separately without involving a data processing or computer terminal. It is simply the use of a telephone line or similar link to transmit data without involving a computer system. The equipment does not immediately make the data collected available for computer processing, that is, the data when sent and/or received are "off-line".

This type of data communication is relatively cheap and simple. Example of this system includes sales data, waybills (documents identifying the destination and content of goods), etc. The principal advantages of off-line transmission are,

- | | | | |
|---------------|------------------|--------------------|----------------|
| (i). Low cost | (ii). Simplicity | (iii). Flexibility | (iv). Security |
|---------------|------------------|--------------------|----------------|

- (2) **Remote Batch Systems:** Batch processing terminals are characterized by large data transfers in two directions without manual involvement. Information from a batch of inventory cards might, be transferred from a warehouse to a remote computer centre, and in return the warehouse would receive an updated inventory list. In some batch applications, large amounts of data flow in one direction only. When a sales representative records sales on a portable computer terminal but waits until end of the workday to transmit the entire day's orders, a large amount of data flows in one direction and little or no data flows in the other direction.

- (3) **On-line Data Collection (without updating):** is the method of using data communications

technology to provide input data to a computer as such input arises – the data are then stored in the computer (say a magnetic disk) and processed either at predetermined intervals or as required. In this case, the transmission device remains linked 'live' to the central computer throughout working hours. Typically, however, transmission is intermittent, data travelling along the telephone line only when some event takes place at the remote location. Application of this system includes Data Logging – this type of system merely collects data on a continuing basis from a monitoring device, which is measuring some action, for example water levels, accumulated numbers etc. The data collected are normally needed for a type of process control system, such as temperature of a steel furnace, recording readings of instruments in a scientific experiment, monitoring the flow of a river etc.

(4) **Enquiry-response Systems:** this system provides the facility for a user to extract information from a computer. The enquiry facility is passive, that is, does not modify the information stored. The interrogation may be simple, for example, "RETRIEVE THE RECORD FOR EMPLOYEE NUMBER 1234" Such systems may use terminals producing hard copy and/or visual displays.

(5) **Real-time Systems:** are those in which information is made available to and processed by a computer system in a dynamic manner so that either the computer may cause action to be taken to influence events as they occur (for example as in a process control application) or human operators may be influenced by the accurate and up-to-date information stored in the computer, for example in airline reservation systems.

Uses of Data Communications:

In Table 1, a list of five types of data communication systems is given. This table also summarizes many important characteristics of these typical uses of data communications, giving specific application examples and typical characteristic for each application.

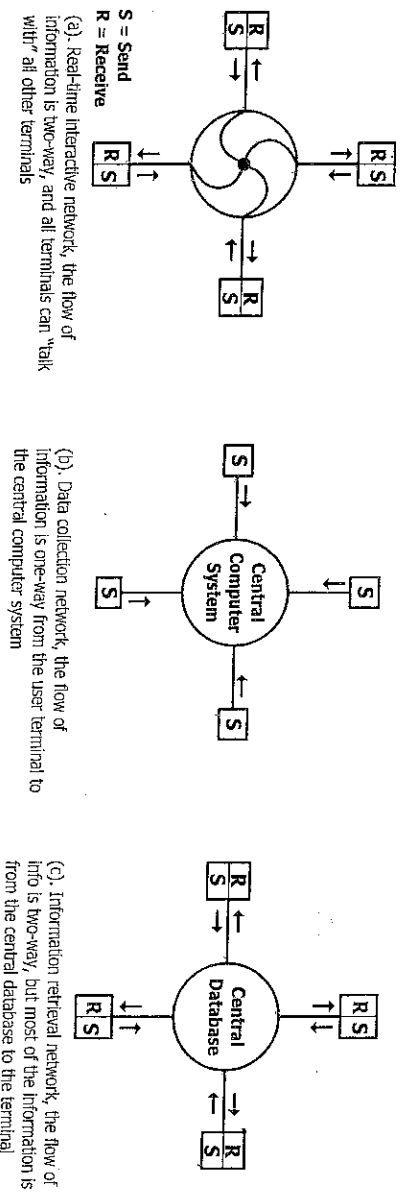
Table 1: Typical Use and Application of Data Communication Systems

Type of Data Comm. System	Used For	Examples of Applications	Typical Characteristics
1. Off-line Data Transmission	Data entry and Data collection	Sales status data. Inventory control. Payroll data gathering.	Data is collected several times per day or week. Direct reply to message not issued for immediately.
2. Real-time Systems	Real-time interactive data entry, retrieval and updating.	Point-of-sale system Airline reservations. Electronic Data Interchange (EDI)	Data arrive frequently (in few secs) and demand response in a few secs.
3. Remote Batch	Remote Job Entry (RJE) and Data Collection.	Remote high speed reading and printing. Local access to distant computer power.	Data is usually bunched and require processing times ranging from minutes to hours. Input and output for each transaction may take seconds or minutes.
4. Enquiry-response System	Information Retrieval.	Credit checking. Bank account status	Low data input per transaction, response

	Hospital information systems.	required within seconds. Output message lengths usually short.
5. On-line Data Collection (without updating)	Real-Time Data Acquisition and Process Control.	Numerical control of machine tools. Remote meter and gauge reading. Remote sensors continuously sampled and monitored at widely varying time intervals.

Flow of Information in Data Communication Network

Figure 2 shows the flow of information in three different types of networks: – those requiring real-time interaction between terminals, those designed for collecting data, and those used for retrieving information.



1.5 FUNCTIONS OF EQUIPMENTS USED IN DATA COMMUNICATION

(a) Line Controllers and Line Drivers:

Line controllers are used to control the flow of data across a physical medium to ensure that information is transferred from source to sink efficiently e.g. remote intelligent terminal controller.

Line Drivers are used to amplify the signal being passed across the communication channel thereby extending the transmission distance between the terminals connected in the network. Line drivers put more electrical power (amperage) on the communication line to drive the information signal for a distance greater than several kilometres. Line drivers are used to connect point-of-sale (POS) terminals, sensors and many of other digital devices to a host computer. A short haul modem is a typical example.

(b) DATA SETS (Modems and DSU/CSU):

These are devices used to facilitate the transmission of encoded information (data) using either analog or digital transmission techniques. Data sets include modems, digital service unit / channel service unit (DSU/CSU).

Modem: - Modem is an acronym for **MOD**ulator/**DEM**odulator. A modem is an electronic device that connects computers via a telephone line, allowing the exchange of information. It consists of a modulator to convert computer information (digital signal) into a telephone signal (analog signal) for transmission on an analog communication channel (e.g. telephone line or voice grade circuit) and a demodulator to convert it back again into digital signal at the receiving end.

multiplexed signals out from the transmission line and directs them to their proper recipient. Figure 3 shows a typical four level multiplexed circuit.

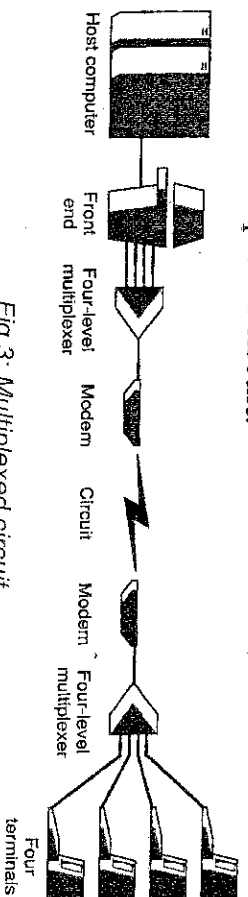


Fig 3: Multiplexed circuit

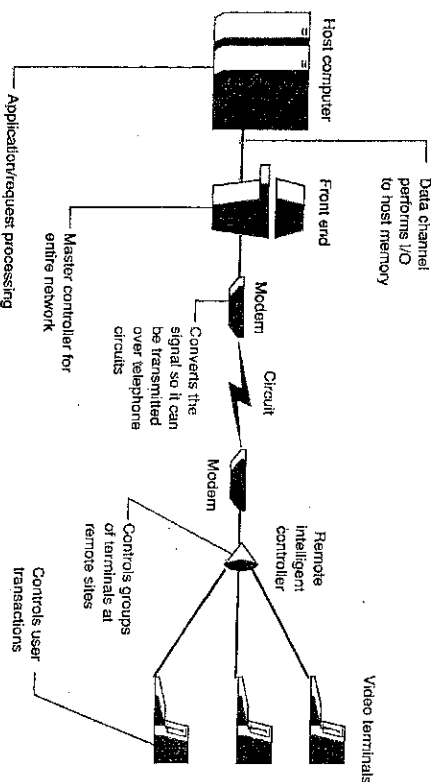
Concentrator: - This is a type of multiplexor that combines multiple channels onto a single high speed transmission channel in such a way that all the individual channels can be simultaneously active. Unlike a multiplexer a concentrator collects data in a temporary storage area whereas a multiplexer spreads the signals back out again on the receiving end.

Modern concentrators are special forms of statistical multiplexers. They are used for the same purposes as multiplexers, except that they are programmed (intelligent) statistical multiplexers.

OTHER EQUIPMENTS: - These equipments are used for network control, network monitoring and network diagnostic and testing.

Front End Processors (FEPs)

A front-end processor is a dedicated small-sized computer interposed between a host computer and a number of networks, such as number of peripheral devices, like terminals, disk units, printers, scanners and tape units.



The function of FEP is to handle the task of managing and communicating with peripheral devices for the host computer such as transmitting and receiving messages, multiplexing, protocol conversion, packet assembly and disassembly, error detection, and error correction e.t.c.

MONITORING DEVICES: - These devices are used to monitor and evaluate the performance of data communication systems. Examples are described below:

Response Time Analyzer checks the operation of the communication protocols in a polled network. It measures the amount of time that elapses from the initiation of polling until receipt of the response.

Data Line Monitor: - A data line monitor does not only performs response time analysis but it also checks the actual data (both control and data characters) on the communication circuit.

Protocol Analyzers measure the responses of all hardware in a network. This equipment can determine whether the network equipment is meeting specifications. These devices also allow you to see the content (bits) of a frame packet during its transmission.

DIAGNOSTIC AND TESTING EQUIPMENTS: - These are equipments used to test the performance and diagnose faults across the data communication system. Descriptions of the various types of test equipment follow:

Breakout Box: - It is a handheld device that can be plugged into a modem's digital side to determine the voltage values for the RS232 connector cable interface (25-pin cable), the V.24 interface, and others.

Bit-Error Rate Tester (BERT): - The Bit-Error Rate Tester is a somewhat more sophisticated piece of digital test equipment than a breakout box because it sends a known pseudorandom pattern over the communication circuit. When this pattern is reflected back, the BERT compares the transmitted pattern with the received pattern and calculates the number of bit errors that occurred on the communication circuit.

Bit-Error Rate (BER) measurements can be made with this type of equipment. A BER is the number of bits received in error divided by the total number of bits received. Service personnel use BER measurements to tune and evaluate the quality of the communication channel.

Block-Error Rate Tester (BKERT): - The Block-Error Rate Tester calculates the number of received blocks that contain at least one bit error divided by the total number of blocks received; this is Block-Error Rate. Assume a BKER measurement has been made and the BKER value is 10^{-2} ($\frac{1}{100}$). This means that out of every 100 blocks received, one contained an error; therefore you would expect to see one retry for every 100 blocks transmitted (1% percent error rate).

Fibre Identifier: - The purpose of a fibre identifier is to locate a particular nonworking fibre without interrupting service on fibre optic network. (A fibre cable may contain a bundle of 72 or 144 glass fibres).

Cable Analyzer: - The cable analyzer is a tester that checks LAN twisted pair wires for signal continuity, pulse distortion, parity, conductivity, connectivity, polarity, reversals, and excessive noise in the data stream.

EXAMPLE 1. The time taken to transfer 100 employee records to corporate headquarters is 1 hour and each record contains 36 MB of data. If each byte was encoded using 8 bit ASCII code plus 2 extra bits for start-stop transmission, determine the speed of the modem used.

SOLUTION

(i) **GIVEN:** No. of Records = 100

Number of Bytes per record = 36 MB = 36,000,000 bits

Download time = 1 hour = $1 \times 60 \times 60 = 3600$ seconds

No. of Bits per byte = $8 + 2 = 10$ bits

The modem that transmits the digital signal is the modulator because it modulates, or puts some form of intelligence on the carrier wave, whereas the receiving modem is the demodulator because it demodulates or interprets the original digital signal upon receiving it. To modulate is to translate an information signal onto another signal (a carrier) so it can be transmitted over the type of transmission circuits in use. The modem performs this adaptation.

The speed of a modem determines how fast it can send and receive information through telephone lines. Modems transmit data at different speeds, measured by the number of bits of data they send per second (**bps**). A 28.8 kbps modem sends data at 28,800 bits per second. A 56 kbps modem is twice as fast, sending and receiving data at a rate of 56,000 bits per second. Faster modems transfer information more quickly. The speed at which information is transferred also depends on the quality of the phone line/link/channel. For example, a modem with a speed of 28.8 kbps may not reach that speed if the phone line quality is poor. A separate phone line is required to use a modem.

Digital Modem (DSU/CSU): - If the communication system use digital transmission for their entire length (e.g. ISDN or T-1 line), instead of analog transmission as is done with normal telephone channels, a digital modem is used. The function of DSU/CSU is to convert a digital signal from a computer to a more precise and more accurate digital signal. For example, a digital modem can take a weak electrical signal and shape the pulses, put very precise timing characteristics between the pulses, increase the strength, and control its electrical characteristics. This is done to reduce noise, distortion and errors. The correct name for this modem is either a Channel Service Unit (CSU) or Digital Service Unit (DSU).

An Integrated Services Digital Network (ISDN) carries digital signals throughout the transmission process. Hence, a digital modem would be required for this type of transmission line to carry information. This is because a digital modem does not convert between digital and analog signals; it processes or formats the digital signal between the computer and the ISDN lines.

(c) TRAFFIC CLUSTERING DEVICES (Multiplexers and Concentrators)

Traffic clustering devices are used to avoid data traffic collisions. These devices provide availability and scalability of a data communication system. They provide improved performance of such systems. Some features and benefits of these devices include: load balancing, fail over, fault resilience, multi-port disk access, and scalability. Some equipment used to achieve these includes multiplexers and concentrators.

Multiplexers: This is a device that combines data traffic from several low speed communication circuits onto a single high speed transmission line. Depending on the multiplexer 32 or more devices share a single communication line.

To multiplex is to place two or more simultaneous messages on a single transmission line. Multiplexing on a data communication network means that two or more messages are sent simultaneously over one communication circuit. Messages sent by a multiplexer must be received by a multiplexer of the same type. The receiving multiplexer i.e. **demultiplexer** separates the

$$\text{Generally, File Transfer Time (s)} = \frac{\text{No. of Records} \times \text{No. of Bytes} \times \text{No. of Bits per byte}}{\text{Transmission Speed (bps)}}$$

$$\begin{aligned} \text{Therefore, Txm Speed (bps)} &= \frac{\text{No. of Records} \times \text{No. of Bytes} \times \text{No. of Bits per byte}}{\text{File Transfer Time (s)}} \\ &= \frac{100 \times 36,000,000 \times 10}{3,600} \\ &= 10,000,000 \text{ bps} \\ &= \mathbf{10 \text{ Mbps}} \end{aligned}$$

BYTE: A byte (B) is the standard size for digital data or encoded information in modern data communication systems. It is a unit of data represented by a group of 8 bits. Standard and multiple units of the byte are given in Table 2. For all theoretical calculations the decimal value of is used.

Table 2. Byte and its multiple

Unit Symbol	Base 10	Decimal Value (B)	Binary value (Base 2)	Byte value (Bytes)	Bit value (Bits)
1 B	10^0	1	2^0	-	8
1 kB	10^3	1,000	2^{10}	1,024	8,192
1 MB	10^6	1,000,000	2^{20}	1,048,576	8,388,608
1 GB	10^9	1,000,000,000	2^{30}	1,073,741,824	8,589,934,592
1 TB	10^{12}	1,000,000,000,000	2^{40}	1,099,511,627,776	8,796,093,022,208

EXAMPLE 2. A digital test equipment was used to send the bit pattern – (1 0 0 0 1 1 1 0 0 0) on three different tests over a 1200 bps communication line. The following bit patterns were reflected,

(1) 1 1 0 0 1 1 1 0 0 1 (2) 1 0 0 0 0 0 1 1 0 0 0 (3) 1 0 0 1 1 0 0 0 1 1 0

Calculate the BER in each test.

SOLUTION

$$\text{Bit Error Rate (BER)} = \frac{\text{Number of Bit errors}}{\text{Number of Bits received / sent}} \times 100$$

To solve this, the sent and received bit patterns are compared to get the number of errors.

(i). Comparing the sent and received bits,

✓	1	0	0	0	1	1	1	0	0	0	✓
	1	1	0	0	1	1	1	0	0	1	

The number of errors = 2

$$\text{Therefore, (BER)} = \frac{2}{10} \times 100 = \mathbf{20\%}$$

(2). Comparing the sent and received bits,

✓	1	0	0	0	1	1	1	0	0	0	✓
	1	0	0	0	0	1	1	0	0	0	

The number of errors = 1

Therefore, (BER) = $\frac{1}{10} \times 100 = 10\%$

- (3). Comparing the sent and received bits,

	✓	✓	✓	✓	✓	
1	0	0	0	1	1	0
0	0	0	1	1	0	0
1	0	0	1	1	0	1
0	1	0	0	1	1	0

The number of errors = 5

Therefore, (BER) = $\frac{5}{10} \times 100 = 10\%$

TABLE OF ASCII CODE

Bits 7654321	Character	Bits 7654321	Character	Bits 7654321	Character	Bits 7654321	Character
0000000	NUL	0100000	SP	1000000	@	1100000	`
0000001	SOH	0100001	!	1000001	A	1100001	a
0000010	STX	0100010	"	1000010	B	1100010	b
0000011	ETX	0100011	#	1000011	C	1100011	c
0000100	EOT	0100100	\$	1000100	D	1100100	d
0000101	ENO	0100101	%	1000101	E	1100101	e
0000110	ACK	0100110	&	1000110	F	1100110	f
0000111	BEL	0100111	'	1000111	G	1100111	g
0001000	BS	0101000	(	1001000	H	1101000	h
0001001	HT	0101001	)	1001001	I	1101001	i
0001010	LF	0101010	*	1001010	J	1101010	j
0001011	VT	0101011	+	1001011	K	1101011	k
0001100	FF	0101100	,	1001100	L	1101100	l
0001101	CR	0101101	-	1001101	M	1101101	m
0001110	SO	0101110	.	1001110	N	1101110	n
0001111	SI	0101111	/	1001111	O	1101111	o
0010000	DLE	0110000	0	1010000	P	1110000	p
0010001	DC1	0110001	1	1010001	Q	1110001	q
0010010	DC2	0110010	2	1010010	R	1110010	r
0010011	DC3	0110011	3	1010011	S	1110011	s
0010100	DC4	0110100	4	1010100	T	1110100	t
0010101	NAK	0110101	5	1010101	U	1110101	u
0010110	SYN	0110110	6	1010110	V	1110110	v
0010111	ETB	0110111	7	1010111	W	1110111	w
0011000	CAN	0111000	8	1011000	X	1111000	x
0011001	EM	0111001	9	1011001	Y	1111001	y
0011010	SUB	0111010	:	1011010	Z	1111010	z
0011011	ESC	0111011	;	1011011	[	1111011	{
0011100	FS	0111100	<	1011100	\	1111100	
0011101	GS	0111101	=	1011101	]	1111101	}
0011110	RS	0111110	>	1011110	^	1111110	~
0011111	US	0111111	?	1011111	_	1111111	DEL

EXERCISE 1

- (1) In a Data Communication System distinguish between an on-line system and off-line system.
- (2) A 2 million-byte file encoded using 7-bit ASCII code plus 1-parity check bit is to be downloaded from an Internet website. Determine how long it will take to download this file using;
 - (i). 56 kbps modem
 - (ii). 28.8 kbps modem

Comment on your answer in (i) and (ii) above.
- (3) In one sentence only explain why a modulator and demodulator are combined into a single piece of device (i.e. the modem)?
- (4) Do you need an analog modem for an all-digital network? Explain.

MODULE TWO

MODULATION AND MULTIPLEXING

2.1 MODULATION

Modulation is a technique used by modems to transmit information by transforming the information (digital or analog) into analog carrier signals which can be transmitted over a transmission medium without distorting the signals. This technique tries to modify the form of an electrical signal so that the signal can carry information on a communication medium. The modulated signal is referred to as an analog signal. The signal that does the carrying is the CARRIER WAVE. Modulation changes the shape or form of the carrier wave to transmit '0's and '1's.

MODULATION can be defined as the process by which a property (e.g. amplitude, frequency and phase) of a high frequency signal (known as CARRIER) is varied in proportion with the instantaneous value of the message (MODULATING) signal.

In communications, a CARRIER WAVE is a continuous oscillating voltage of constant amplitude, frequency and phase and it carries no information, but, it can be altered by a series of pulses (bits in coded form), which can alter either the amplitude (height), frequency or phase. When this occurs the carrier has been modified to carry information. The modified carrier wave is called modulated signal. The equipment in which modulation takes place is called a modulator. If the modulator caused the amplitude of the carrier wave to vary, the result is called Amplitude modulation and so on for frequency and phase.

The original digital signal sent by a computer is usually suitable for transmission in a voice-grade channel (telephone line). A pair of modems is used to generate a carrier in which the digital data are translated onto this carrier. The carrier signal is a modulating analogue signal at a specific frequency. To retrieve the modulated information, the receiving modem tunes to the carrier signal's frequency and extracts the information. The recovery of the transmitted information from the carrier signal is known as DEMODULATION. Modems can perform either analogue modulation or digital modulation technique for either analogue or digital transmission systems.

2.2 NEED FOR MODULATION

Modulation technique is employed in communication systems for the following reasons:

- (1) For Channel Transmission: The message has to be translated in a form suitable for transmission over a physical medium.
- (2) For Channel Assignment: Each message signal is transmitted at a unique frequency band to avoid mix-up with other signals. This is why for example; one radio station only is received within its coverage area without interference from other station.
- (3) To reduce noise and interference.

- (4) For multiplexing so that several messages can be transmitted through a single channel.
- (5) For easy radiation and reception of signals using practically realizable antenna sizes.
- (6) To overcome equipment limitations e.g. size and weight.

2.3 TYPES OF MODULATION

Modulation can be grouped into two categories, (i). Analogue Modulation (ii). Digital Modulation

ANALOGUE MODULATION:

There are THREE basic analogue modulation techniques which uses the physical property of an analogue carrier signal (sine wave) to carry information, these are:

- (i). Amplitude Modulation (AM) (ii). Frequency Modulation (FM) (iii). Phase Modulation (PM)

Amplitude Modulation (AM): This is a modulation technique in which the amplitude (height) of the carrier signal is varied in accordance with the instantaneous value of the message signal (data) that is to be transmitted. When the message signal (data) to be transmitted is digital (0s and 1s), the technique is called Amplitude Shift Keying (ASK).

The amplitude shown in figure 5(c) depicts the peaks at the highest amplitude representing binary 1s and the peaks at the lowest amplitude representing binary 0s.

Amplitude modulation is suitable for data transmission, and it allows efficient use of the available bandwidth of a voice grade line (telephone line). However, it is more susceptible to noise (more errors) and fading during transmission than frequency modulation because the information is contained in the change of amplitude which is easily affected by noise.

Frequency Modulation (FM): This is a modulation technique in which the frequency of the carrier signal is varied in accordance with the instantaneous value of the message signal. When frequency modulation is used to transmit digital signals, it is called Frequency Shift Keying (FSK).

In FSK, each 0 or 1 is represented by a different frequency. In this case the amplitude does not vary. Figure 5(d) shows frequency modulation as a bit stream of 0s and 1s. Notice that the amplitudes are equal. FM is less affected by noise and fading because the information is contained in the change of frequency not amplitude.

Modems employing FSK may use three basic frequencies, such as 1700 Hz, 1200 Hz, and 2200 Hz. In this case when no data is being transmitted, the carrier wave signal is 1700 Hz. When a stream of 0s and 1s is transmitted, the carrier wave switches between 1200 Hz and 2200 Hz depending on whether it is a 0 or 1 that is transmitted. FSK modems are wasteful in their use of bandwidth since each data signal is transmitted at a unique frequency, with the result that the maximum data rate over normal telephone lines is 1800 bps.

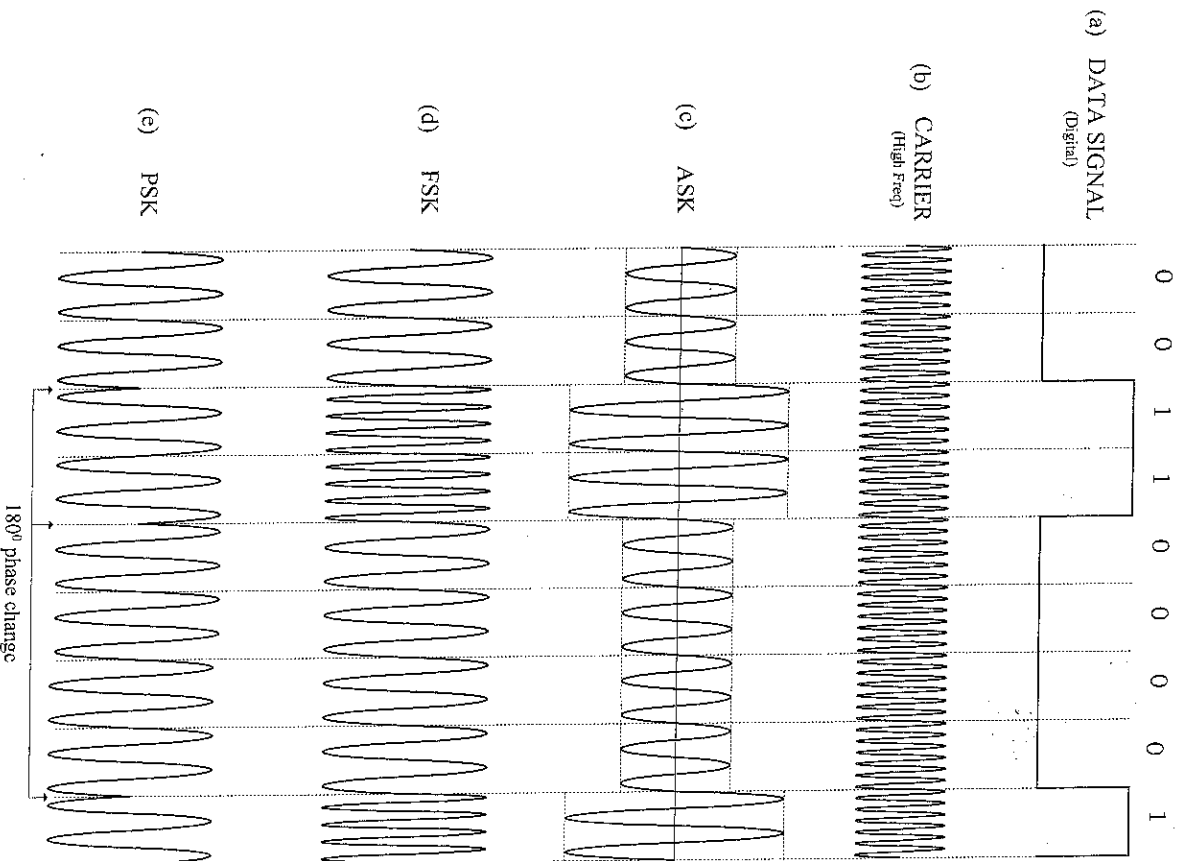


Fig. 5 Amplitude, Frequency and Phase Modulation

Phase Modulation (PM): This is a form of modulation in which the phase (cycle) of the carrier signal is varied in accordance with the instantaneous value of the message signal. During phase modulation the cycle of the wave is shifted by $\frac{1}{2}$ of its cycle. It is this shift in the cycle that is used to represent the binary values of a digital signal; this technique is called Phase Shift Keying (PSK).

Figure 5(e) shows a Two-phase (0° and 180°) PSK for a stream of 0s and 1s. Notice that every time there is a change in state (0 or 1); there is a 180° change in the cycle. In 180° phase shift, the sine wave immediately goes in the other direction.

Its main advantage over FSK is that as it uses a single frequency it uses much less bandwidth. It is thus less affected by noise contained in the carrier frequency and has an advantage over ASK because its information not contained in the amplitude of the carrier, thus again it is less affected by noise.

EXAMPLE 1.

A data communication channel having a bandwidth of 53 kHz is to be used to combine a number of sub-channels each of freq range 400 Hz - 4.4 kHz. If a total of 5 kHz is used for guardband, determine,

- (1) the bandwidth of each sub-channel
- (2) the number of sub-channels that can be combined.
- (3) the number of guardbands.
- (4) the bandwidth each guardband.
- (5) Sketch a typical arrangement of the sub-channels and guardbands.

SOLUTION

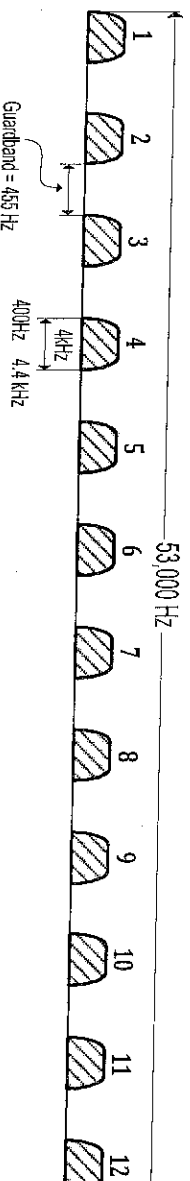
Total bandwidth of the channel = 53 kHz = 53,000 Hz

- 1) Bandwidth of each sub-channel = $4,400 - 400 = \mathbf{4,000\text{ Hz}}$

Total BW reserved for guardband = 5,000 Hz

Total BW to be used for sub-channels = $53,000 - 5,000 = 48,000\text{ Hz}$

- 2) Therefore, number of sub-channels to be combined = $\frac{48,000}{4,000} = \mathbf{12\text{ sub-channels}}$
- 3) Number of guardbands for 12 sub-channels = $12 - 1 = \mathbf{11\text{ guardbands}}$
- 4) Bandwidth of each guardband = $\frac{5000}{11} = \mathbf{455\text{ Hz}}$
- 5) Typical arrangement given below,



EXERCISE 2A

Five channels with 100 kHz bandwidth are to be multiplexed together. What is the minimum bandwidth of the channel if there is need for a guard band of 10 kHz between the channels to prevent interference?

2.4 MULTIPLEXING

Communication channels always have far greater capacity (bandwidth) for data transmission than a single device or computer can use. Communication channels always have far greater capacity (bandwidth) for data transmission than a single device or computer can use. The rationale for multiplexing is due to the high cost of operating communication channels which makes it more efficient if several devices can be connected to share the bandwidth of the channel at the same time.

Multiplexing is the transmission of multiple message signals at the same time over a single communication channel. Whenever the bandwidth of a communication channel linking two devices is greater than the bandwidth needed by the devices connected to the channel, the channel can be shared (more devices can be connected to the channel). If the bandwidth of a channel is greater than the bandwidth required by the devices connected to it, the bandwidth is wasted.

Devices that are used for multiplexing are: (i) Multiplexers (ii) Concentrators

(iv) Modems

(iii) FEPs

MULTIPLEXER: A communications device that multiplexes (combines) several signals for transmission over a single communication channel. Depending on the multiplexer up to 32 devices can share a single communication line. Information sent by a multiplexer must be received by a demultiplexer of the same type. A DEMULTIPLEXER separates the individual messages from the transmission line and directs them to the proper destination. Frequently a multiplexer and demultiplexer are combined into a single device capable of processing both outgoing and incoming signals. The advantages offered by using multiplexing include:

- (i) Channel shared by multiple devices
- (ii) Reduced line costs
- (iii) Reduced modem costs
- (iv) Increased line utilization
- (v) Transparent - this configuration is invisible to the users.

An important aspect of multiplexing is Transparency. Transparent means the hardware multiplexer box does not in any way interrupt the flow of data. Neither the computer, nor the modem, nor the terminal/microcomputer using the modem knows the multiplexer is being used regardless of the type of transmission circuit used. When the circuit is multiplexed at one end and demultiplexed at the other, each user's terminal thinks it has its own direct connection to the host computer.

Multiplexing usually is done in multiples of 4, 8, 16, and 32 simultaneous transmissions over a single communication circuit. Multiplexers can be separated into three major categories as:

- (i). Frequency Division Multiplexing (FDM)
- (ii) Time Division Multiplexing (TDM)
- (iii) Statistical Time Division Multiplexing (STDM)

Frequency Division Multiplexing (FDM).

Frequency division multiplexing can be described as a broadband (analog) transmission technique where multiple signals are modulated at different frequencies and transmitted over a single transmission channel simultaneously. In FDM, signals generated by each sending device modulate different carrier frequencies. These signals are then combined into a single composite signal that can be transmitted by the channel. It can be used when the bandwidth of the channel is greater than the combined bandwidth of the signals to be transmitted. It was used in early telephone systems.

In FDM, the multiplexer uses the 3,000 Hz of available bandwidth of a telephone channel (voice grade circuit), dividing it into multiple sub-channels. The multiplexer in Fig. 6 subdivides the bandwidth of the channel into four pairs of frequencies to allow four simultaneous transmissions of 'o's and 'i's.

The guardbands in figure 6 are the unused portion of bandwidth that separates each pair of frequencies from the other. They keep the signals in each of the four sub-channels from interfering with the adjacent sub-channels and allow space for frequency drift. With FDM, the frequency

multiplexer and the modem are usually combined into a single piece of hardware. For example multiport modems are frequency division multiplexers.

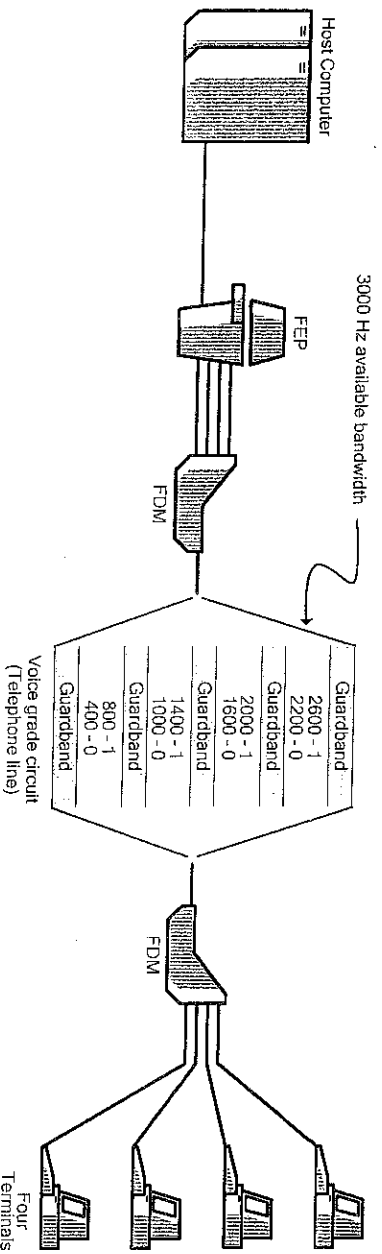


Fig. 6: Frequency Division Multiplexed Circuit (FDM)

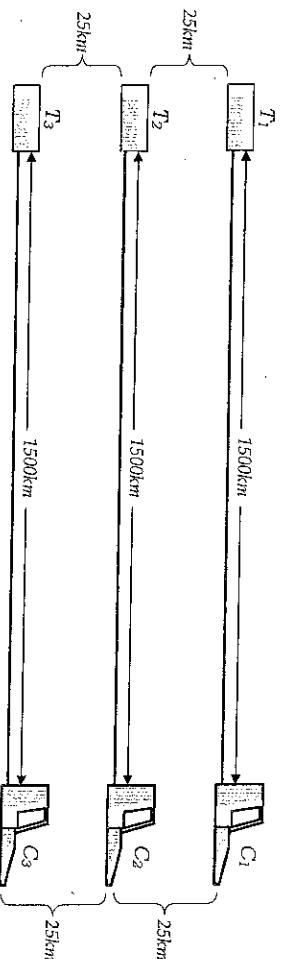
Another characteristic of FDM is that the sub-channels need not all terminate at the same location as they do in figure 6. Multi-drop means the four terminals in figure 6 can be at separate locations, such as on different floors in a building or even at different locations in a city. Therefore frequency division multiplexing can be used in a multi-drop network where each drop-off to a terminal operates at a different frequency and each terminal is in a different location.

Frequency division multiplexers are somewhat inflexible because, once you determine how many sub-channels are required; it may be difficult to add more sub-channels without purchasing an entirely new frequency division multiplexer that is divided into greater number of sub-channels. In addition, maintenance cost usually is greater for frequency division multiplexing equipment than for time division multiplexing equipment.

Cost Effectiveness of Multiplexers

One way of reducing the total circuit cost of operating a data communication network is to consider the possibility of acquiring a multiplexer. The primary benefit of multiplexing is to save communication circuit costs between the host computer or business organization and many remote terminals. Figure 9 shows how multiplexing can save on communication costs.

EXAMPLE 2. Three terminals (T1, T2, T3) are to be connected to three computers (C1, C2, C3) so that T1 is connected to C1, T2 to C2, and T3 to C3. All are in different cities. T1 and C1 are 1500km apart, as are T2, C2 and T3 and C3. The points T1, T2, and T3 are 25km apart, and the points C1, C2 and C3 also are 25km apart as shown in the figure below.



If telephone lines cost ₦1 per kilometre,

- (i) What is the total line cost for three independent lines?

- (ii) If a multiplexer/demultiplexer cost ₦2,000, can you save money by another arrangement of the lines? Describe the arrangement.

- (iii) Determine how much is saved by this arrangement.

SOLUTION

- (i). Cost of single telephone line = ₦1/Km x 1,500km = ₦1,500

Total cost of three independent telephone lines = $3 \times \text{₦1,500} = \text{₦4,500}$

- (ii). YES. By using a multiplexer to combine signals from the three terminals into one telephone line at one end, and then using a demultiplexer to separate the individual signals from the line and direct them to the proper computer at the other end.

- (iii). New cost = Cost of multiplexer/demultiplexer + Cost of one telephone line
 $= 2,000 + 1,500 = \text{₦3,500}$

- (iv). Savings = $4,500 - 3,500 = \text{₦1,500}$

Time Division Multiplexing (TDM).

Time division multiplexing is a digital (baseband) transmission technique, where separate time slots are assigned to two or more terminals to use a communication channel. The technique actually divides transmission time by assigning to each device a time slot during which it can send or receive data hence all terminals receive equal share of the channel. During these time slots data may be sent or received without conflicting with data sent from another terminal. Here, instead of dividing a portion of the channel bandwidth (frequency) as in FDM, time is shared.

In TDM, (figure 7) the data flow from each input terminal is divided into units. The multiplexer takes a unit from each terminal and puts them together into a frame. A unit can be 1 bit, one character, or one block of data. If we have n-terminals, a frame is divided into n-time slots and one slot is allocated to each input terminal.

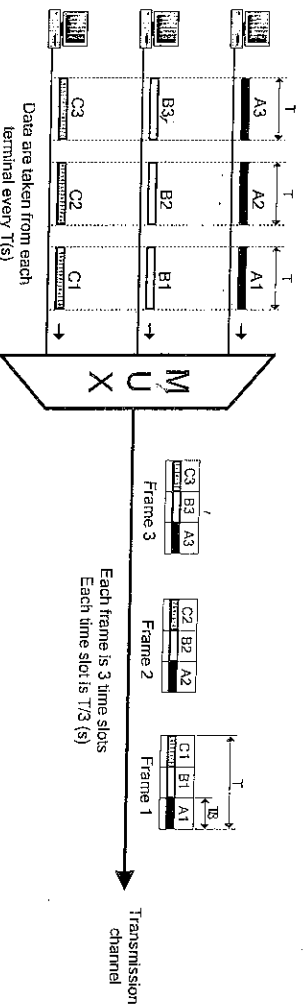


Fig. 7: Framing of bits in TDM scheme

Each input unit occupies one output time slot in the frame. However, the duration of an output time slot in the frame is n-times shorter than the duration of an input time slot. If the input time slot is T (s), the output time slot is T/n (s), and the duration of the frame is T (s), where n is the number of connections.

The data rate (or throughput in bps) of the output channel must be n-times the data rate of each input line to guarantee the flow of data. This means that a unit of data in the output channel has a

shorter duration and travels faster. Figure 8 shows an example of TDM where $n = 3$. The data rate of the channel is 3 times the data rate of each input line. Also, the duration of a unit on each input line is 3 times that of the output time slot.

Time slots are grouped into frames. A frame consists of one complete cycle of time slots, with one slot dedicated to each sending terminal. Even if a terminal has no data to send, the corresponding slot in the output frame is left empty (blank) because it is dedicated to only that terminal.

EXAMPLE 3. In figure 7 above, the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is one bit), what is the duration of:-

- (i) each input slot (ii) each output slot (iii) each frame

SOLUTION

- (i). For each input connection, data rate = = 1 kbps

$$\text{Input bit duration} = \frac{1}{\text{input data rate}} = \frac{1}{1000} = 0.001 \text{ s} = 1 \text{ ms}$$

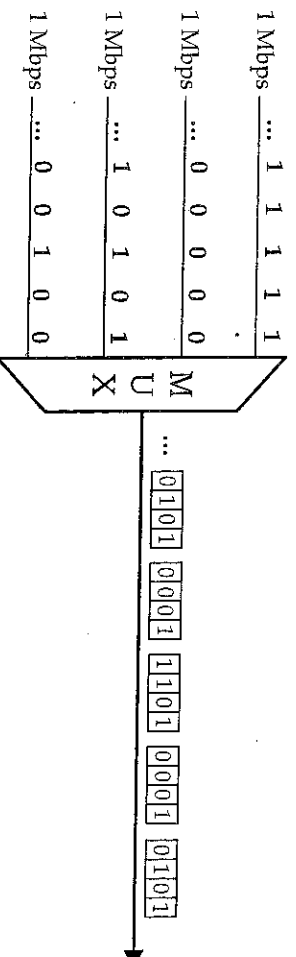
- (ii). Duration of each output time slot = $\frac{T}{3}$, where T is input bit duration and 3 is number of inputs
 $= \frac{1}{3} \text{ (ms)}$

- (iii). Number of output time slots = 3

$$\text{Duration of frame} = 3 \times \frac{1}{3} = 1 \text{ ms}$$

EXAMPLE 4. The figure below shows TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (i) the input bit duration (ii) the output bit duration

- (iii) the output bit rate (iv) the output frame rate



SOLUTION

- (i). Input bit duration = $\frac{1}{\text{input data rate}} = \frac{1}{1 \times 10^{-6}} = 1 \text{ } \mu\text{s}$

- (ii). output bit duration = $\frac{1}{4} \times \text{input bit duration} = \frac{1}{4} \times 1 \text{ } \mu\text{s} = 0.25 \text{ } \mu\text{s}$

- (iii). Output bit rate = $\frac{1}{\text{output bit duration}} = \frac{1}{0.25 \times 10^{-6}} = 4 \text{ Mbps}$ [or $4 \times 1 \text{ Mbps}$]

- (iv). Output frame rate = input bit rate
 $= 1,000,000 \text{ frames per sec}$

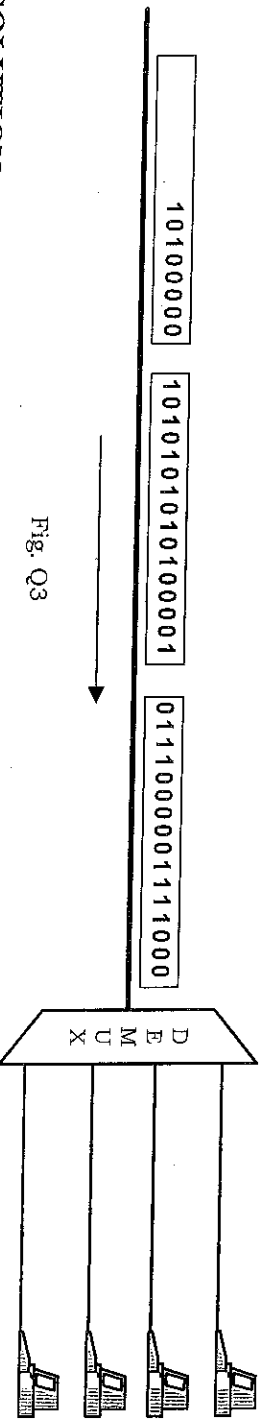
Time division multiplexing generally is more efficient than frequency division multiplexing, but it does require a separate modem. It is easy to expand a time division multiplexer from, let us say, 8 to

12 channels. All TDM channels usually originate at one location and all terminate at another location. Time division multiplexers generally are less costly to maintain than frequency division multiplexers.

Empty Slots

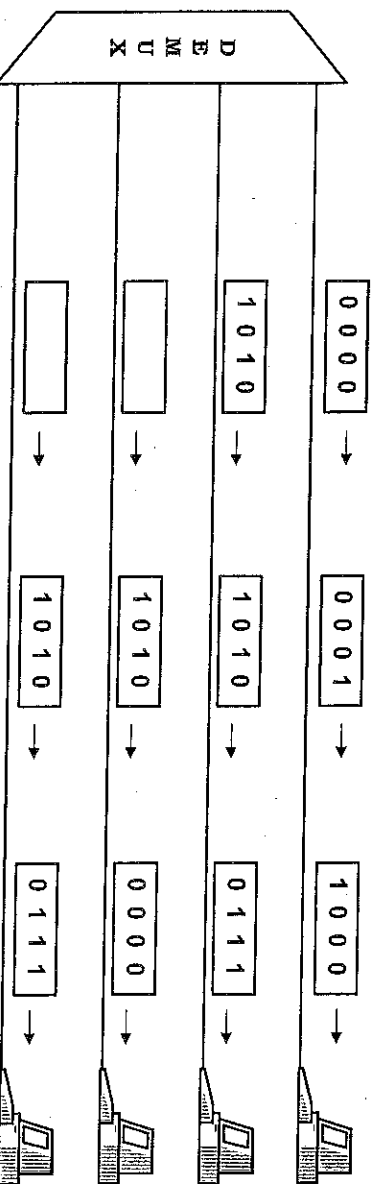
In TDM, if a terminal does not have any data to send, the corresponding slot in the output frame is transmitted empty allowing idle lines to occupy the transmission channel carrying capacity. This makes the technique somewhat inefficient in channel usage. To improve the efficiency, the empty slots (idle lines) have to be removed from the frame. This can be achieved by addressing each character position in the frame through the use of another multiplexing technique – statistical time division multiplexing which is discussed next.

EXAMPLE 5. Fig. Q3 shows a demultiplexer in TDM scheme. If the input is 16 bits long (no framing bits), what is the bit stream in each output? The bits arrive at the demultiplexer as shown by the arrow.



SOLUTION

From the diagram given, there are 4 terminals at the output of the demultiplexer. Each input frame has 16 bits which when divided by the number of terminals means that the demultiplexer will distribute the 16 bits into 4 bits each to each terminal in succession as shown below.



Statistical Time Division Multiplexing (STDM):

Statistical time division multiplexing (also known as stat mux) is a type of TDM technique that provides improved time-sharing efficiency by transmitting data only for those lines with data to send, so that idle lines take up none of the carrying capacity (bandwidth) of the communication channel.

In STDM, the number of slots in each frame is less than the number of input lines. The mux checks each input line in a sequence and allocates a slot for any input line that has data to send; otherwise it skips the line and checks the next line. Figure 8 shows STDM implementation. In this technique, no slot is empty as long as there are data to be sent by any input line.

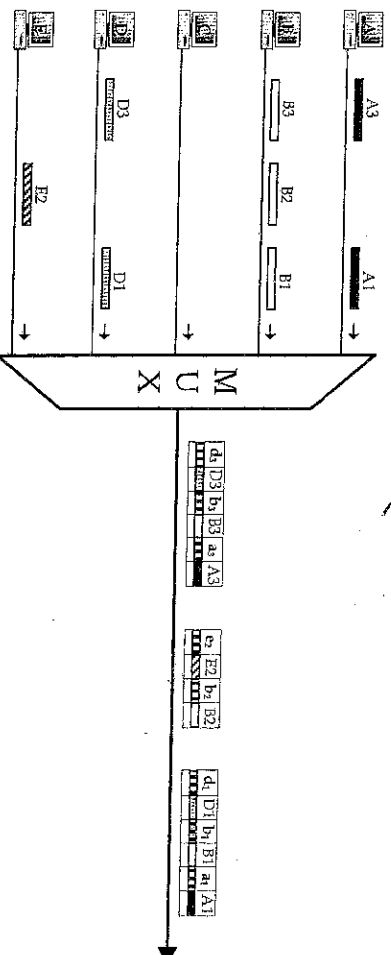


Fig.10: Statistical TDM

Statistical time division multiplexing allows the connection of more terminals to the transmission channel than the capacity of the channel. In its simplest context, if there are 12 terminals connected to a statistical time division multiplexer and each terminal can transmit at 1200 bps, then the total is 14,400 bps transmitted in any given instant of time. However, if the STDM/modem/channel combination has a maximum speed of only 9600 bps, then there might be a period of time when the system is loaded above its capacity.

The STDM technique takes into account the fact that there is some idle time, since not all the terminals will transmit at their maximum rated capacity for every microsecond. Based on this, each character in a frame is addressed and then time division multiplexed on statistical basis. For example in figure 10, the stat mux carries data from a terminal with the address of the destination (receiver) and inserts it into a slot in the frame. The addressing is given by $n = \log_2 N$, where n is equal to the number of address bits, N is equal to the number of output lines. For example for 5 different output lines we need, $n = \log_2 5 \approx 2.32$ or 3-bit address (since a bit cannot have any value between 0 and 1).

The multiplexer takes a character from each terminal only when the terminal has a character to send. The technique is used to check through the 5 terminals, say terminals A, B, D. These are sent immediately. Then all 5 terminals are checked again to determine which terminal need servicing. The process is repeated continuously. At the other end of the channel, the character is given to the proper device because the 3-bit address included with each character identifies the terminal device.

Statistical multiplexers use software and a microprocessor chip built into the multiplexer to support a number of devices at different speeds, without the modem having to equal the total combined speed of all the attached devices.

An important aspect of statistical multiplexing is **FLOW CONTROL**. During peak periods when any of the terminals try to communicate data at their maximum rate, the multiplexer must have a way to stop and start transmission from any of its attached terminals when the terminals try to communicate too much data. Statistical multiplexers use flow control and internal data buffers to send data down a busy channel.

CONCENTRATORS:

A concentrator is a special type multiplexer (or computer) for multiplexing and sharing transmission

lines; it concentrates data travelling to or from a number of remote locations into 'bulk loads' for transmission on a high speed (trunk) transmission line to the host processor.

In today's terminology, concentrators are special forms of statistical multiplexers. Concentrators are used for the same purposes as multiplexers, except that they are programmed (intelligent) statistical multiplexers. However unlike a multiplexer, a concentrator collects data in a temporary storage area. Whereas a multiplexer spreads the signals back out again on the receiving end, the concentrator has a receiving computer to perform that function.

For example, you might concentrate approximately twelve 4800 bps communication channels into one 56,000 bps digital communication channel. Even though this does not work out in perfectly equal way ($12 \times 4800 = 57,600$ bps), the statistical intelligence (program) and memory takes care of the small difference.

Being intelligent, concentrators may be programmed to perform a variety of tasks such as error handling, data validation, data compression, encryption, polling and backup etc. Some concentrators can even switch messages to different communication channels.

In banking ATM environment (see fig 11) where three regional processing centres (Lagos, Abuja, Kano) are responsible for authorizing transactions, each city with multiple ATMs could use a concentrator to handle ATM traffic. The concentrator would have three lines, one each for the three hosts in the three regional processing centres. There would also be one line for each ATM or cluster of ATMs (Adamawa, Bauchi, Gombe, Sokoto, Kebbi and Zamfara States). Based on the customer's ATM card number, the concentrator would switch each transaction to the processing centre closest to the customer's home branch.

Differences between a Multiplexer and Concentrator

As multiplexers took on the additional functions the differences between a multiplexer and a concentrator narrowed. The principal differences between a mux and a concentrator are:

- (1) Concentrators are used one at a time; multiplexers are used in pairs
- (2) A concentrator may have multiple inputs and outputs; a multiplexer has multiple inputs and a single output.
- (3) Concentrators have programmed intelligence and may have auxiliary storage for use in support of an application; multiplexers do not have programmed intelligence
- (4) A concentrator may perform some data processing functions such as device polling and data

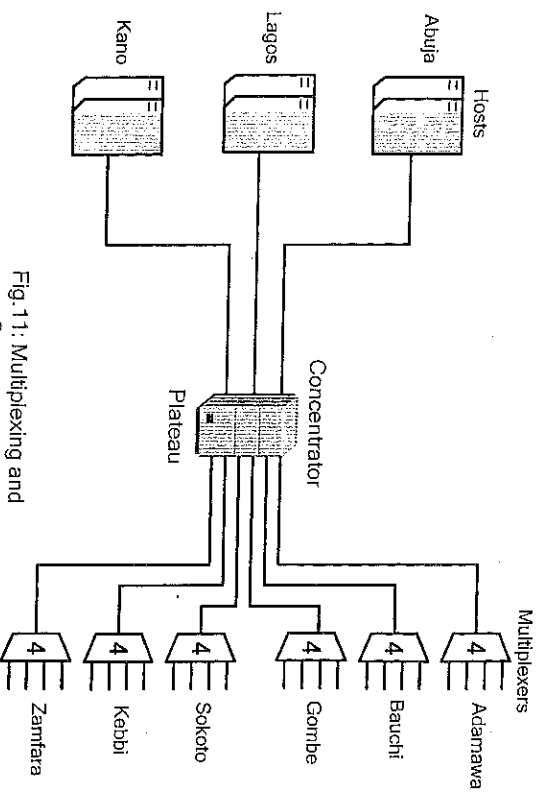


Fig. 11: Multiplexing and Concentrating

validation; multiplexers generally do not process data.

- (5) Multiplexers generally have end user terminals attached to them, while concentrators do not.

EXERCISE 2B

- (1) Mention any 3 differences between TDM and STDM
- (2) Assume a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guardband of 500Hz using FDM. Calculate the required bandwidth.
- (3) We need to use TDM and combine 20 digital sources, each of 100 kbps. Each output slot carries 1 bit from each digital source, but one extra bit is added to each frame for synchronization. Answer the following questions:
 - (i) What is the size of an output frame in bits?
 - (ii) What is the output frame rate?
 - (iii) What is the duration of the output frame?
 - (iv) What is the output data rate?
 - (v) What is the efficiency of the system (ratio of useful bits to the total bits)